



FROM DATA TO BUSINESS DECISIONS

ApexPlanet Software Pvt.
Ltd.

60-Day Data Analytics
Internship — Task 4

BUSINESS PROBLEM

Business Problem

The analysis of the sales dataset revealed differences in:

- Product and category performance
- City-level sales
- Customer purchasing behavior
- Monthly sales trends
- Quantity and pricing patterns

Key Business Question

What factors are associated with higher sales, and how can the business use these insights to improve sales performance?

DATASET & ANALYSIS JOURNEY



Analysis Journey

Raw Dataset



Data Cleaning & Wrangling



Exploratory Data Analysis



SQL Business Analysis



Interactive Dashboard



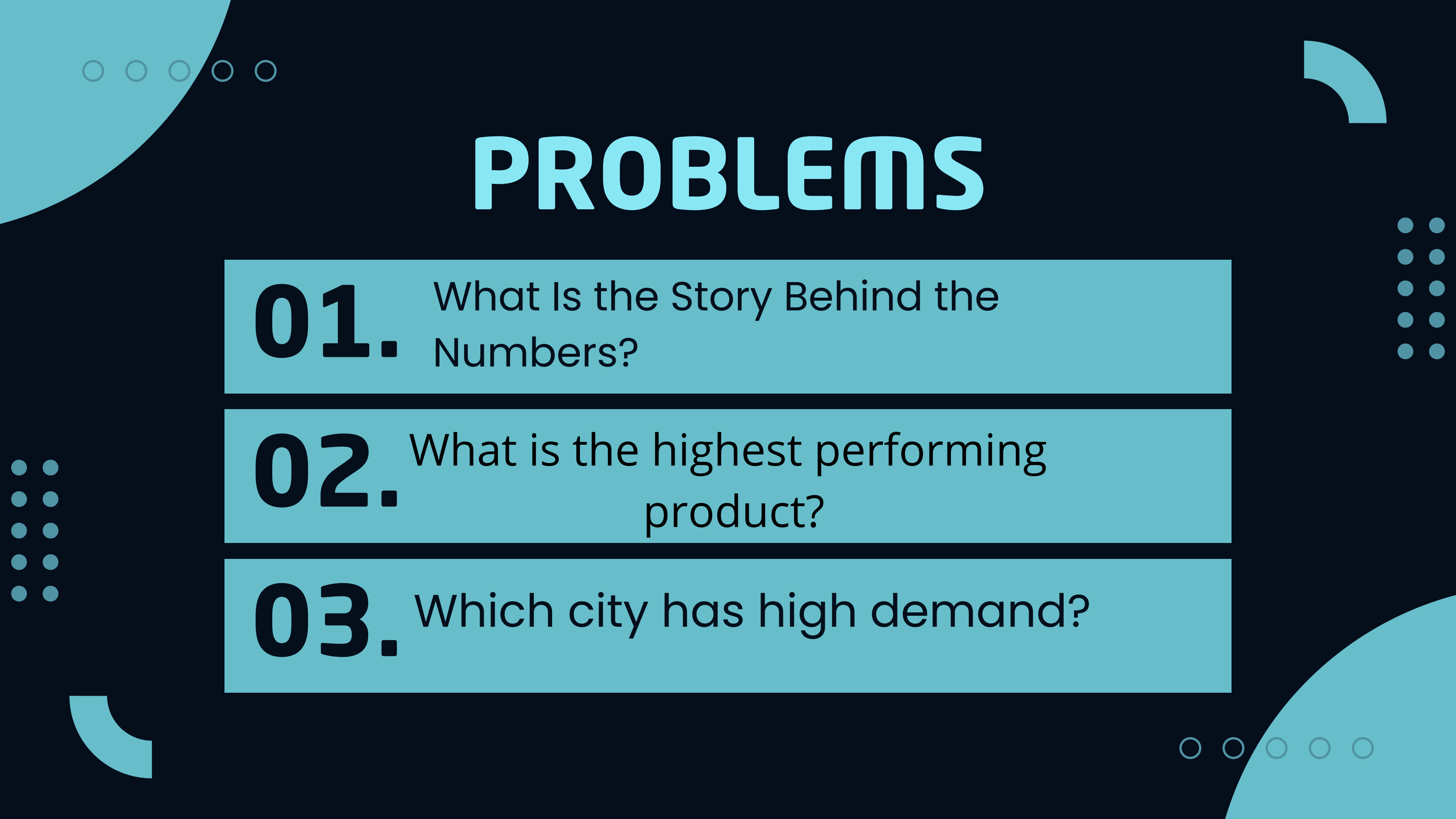
Statistical Validation



Business Recommendations

Dataset

- 1,000 transaction records
- 12 variables
- Customer information
- Order information
- Product information
- Pricing and sales information



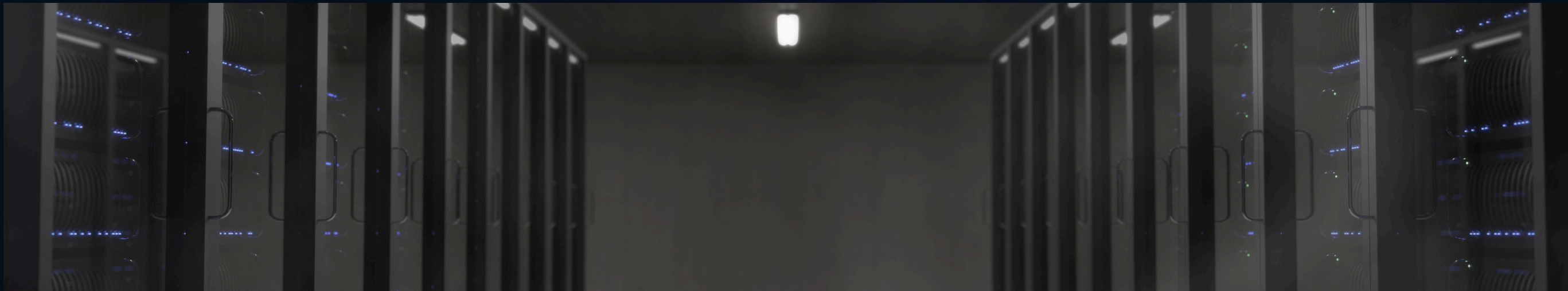
PROBLEMS

01. What Is the Story Behind the Numbers?

02. What is the highest performing product?

03. Which city has high demand?

THE BUSINESS STORY



What Is the Story Behind the Numbers?

The analysis suggests that sales performance is influenced more strongly by transaction characteristics such as:

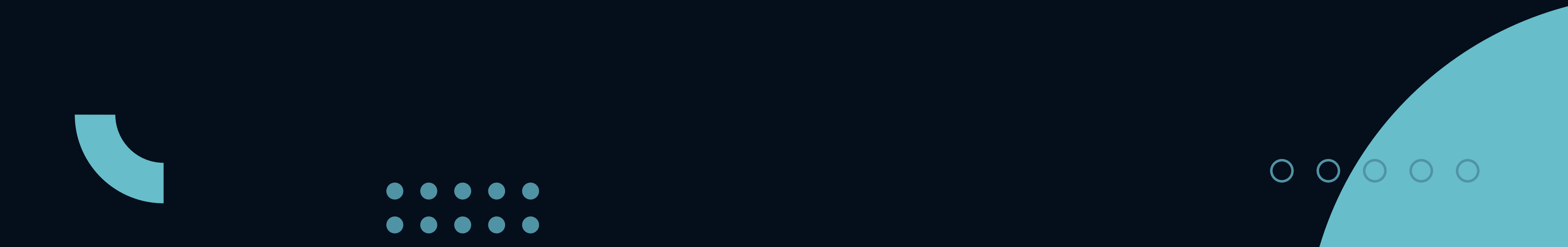
- Quantity purchased
- Unit price
- Product selection
- Category
- Location

rather than simply customer demographics such as age.



BUSINESS INTERPRETATION

The company can potentially improve revenue by focusing on product mix, pricing strategy, inventory availability and high-performing markets rather than relying heavily on demographic targeting alone.



STATISTICAL ANALYSIS

Results of Hypothesis Testing

Pearson Correlation

$r \approx 0.647$

This indicates a moderate-to-strong positive relationship between Quantity and Total Sales.

Interpretation

As the quantity purchased increases, transaction-level sales tend to increase as well.

Statistical Significance

The hypothesis test should be reported using the p-value calculated from your Python analysis.

Final Decision

If your calculated p-value < 0.05 :

BUSINESS IMPLICATIONS

What Does This Mean for the Business?

1. Encourage Larger Baskets

Businesses can explore offers such as:

- Bundle pricing
- Buy-more promotions
- Cross-selling
- Product combinations

2. Focus on High-Performing Products

Products such as laptops and other strong performers can receive greater marketing and inventory attention.

3. Optimize Pricing

Since Unit Price has a relatively strong relationship with Total Sales, pricing decisions should be analyzed alongside demand and quantity.

4. Focus on High-Performing Markets

Cities with stronger sales performance can be studied to identify successful customer and marketing strategies.

CONCLUSION

The analysis transformed raw transaction data into actionable business insights.

Key Takeaways

- Electronics is the strongest-performing category.
- Laptop is the leading product by sales.
- Patna is the highest-performing city.
- Quantity has a positive relationship with Total Sales.
- Unit Price also shows a strong positive relationship with Total Sales.
- Age has almost no linear relationship with Total Sales.
- Statistical validation can help distinguish meaningful relationships from patterns that may occur by chance.

The background is a dark navy blue. It is decorated with various teal-colored geometric elements: circles of different sizes, squares, and curved segments. Some of these shapes are solid, while others are composed of small dots. The text "THANK YOU" is centered in a bold, white, sans-serif font.

THANK YOU